WinPreview

Apple Preview Reverse-Engineered for Windows 11

A native, ultra-lightweight, 60-120fps omnivore document viewer.

Key Architectural Features

1. Bounded LRU Tile Cache (< 200 MB memory ceiling)
2. Google PDFium Native Rendering Engine via Docnet.Core
3. Hardware WIC Image Pipeline (HEIC, RAW, TIFF, PNG, JPEG)
4. 3D Model Viewport (USDZ, glTF, OBJ, STL)
5. ISO 32000-1 Compliant PDF Annotation Interoperability

macOS Preview vs. WinPreview

macOS Preview: 3-8 GB RAM on large PDFs due to unevicted tile buffers.

WinPreview: Strict 200 MB budget with ArrayPool recycling and LRU pruning.